

Yixiu Xiao

PhD Candidate in Mathematics, SJTU
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Personal details

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Education

Apr 2025–present **Visiting Research Student (Joint Training Program)**, *The University of New South Wales (UNSW Sydney)*, Sydney, Australia
Supervisor: Prof. Igor E. Shparlinski.

Sep 2021–expected Mar 2027 **PhD Candidate in Mathematics**, *Shanghai Jiao Tong University*, Shanghai, China
Supervisor: Prof. Hongze Li.

Sep 2017–Jun 2021 **BSc in Mathematics and Applied Mathematics**, *Northwest University*, Xi'an, China

Research Interests

Analytic Number Theory

- Exponential sums, character sums, and bilinear forms
- Kloosterman and Salié sums
- Sieve methods, large-sieve inequalities, and the distribution of primes
- Equidistribution modulo one and discrepancy theory
- Diophantine equations, the Hardy–Littlewood circle method, and the delta method

Current Research Directions

- Squarefree and almost-prime values of norms of $\mathrm{SL}_2(\mathbb{Z})$ matrices
- Sign changes of Kloosterman sums over square-free almost-prime moduli
- Moment estimates and discrepancy for sums of square roots modulo one
- Large-sieve inequalities for quadratic character sums over short intervals
- Shifted bilinear sums of Salié sums and modular square roots of shifted primes
- Least non-trivial zeros of cubic forms and systems of additive cubic equations

Publications

Published

Yixiu Xiao and Hongze Li. **Least zero of pairs of additive cubic equations**. *Forum Mathematicum* 38 (2026), no. 5, 1569–1598. DOI: 10.1515/forum-2025-0367.

Yixiu Xiao and Hongze Li. **Least Zero of a Cubic Form**. *Frontiers of Mathematics* 21 (2026), no. 5, 1043–1069. DOI: 10.1007/s11464-024-0113-6.

Preprints (arXiv)

Yixiu Xiao and Hongze Li. **Selberg sieve weights and sign changes of Kloosterman sums. II. Square-free moduli with at most four prime factors**. *arXiv*: 2609.24255.

Yixiu Xiao and Hongze Li. **Selberg sieve weights and sign changes of Kloosterman sums. I. Uniform asymptotics for shifted weights**. *arXiv*: 2609.24248.

Igor E. Shparlinski and **Yixiu Xiao**. **Arithmetic structure of L_2-norms of $\mathrm{SL}_2(\mathbb{Z})$ matrices**. *arXiv*: 2609.21460.

Yixiu Xiao. **Kloosterman sign changes with moduli having at most five prime factors**. *arXiv*: 2607.10862.

Yixiu Xiao. **Moment Estimates and Discrepancy for Sums of Square Roots Modulo One**. *arXiv*: 2606.28986.

Marc Munsch, Igor E. Shparlinski, Yu-Chen Sun, and **Yixiu Xiao**. **Large sieve inequality for sums of Legendre symbols over short intervals**. *arXiv*: 2604.23661.

Igor E. Shparlinski and **Yixiu Xiao**. **Shifted bilinear sums of Salié sums and the distribution of modular square roots of shifted primes**. *arXiv*: 2601.10113.

Scholarships and Funding

2026 **PhD Support Scheme**, School of Mathematics and Statistics, UNSW Sydney.

2024 **China Scholarship Council (CSC) Scholarship.** Support for 24 months of visiting doctoral research in Australia, April 2025–March 2027.

Talks

25 Aug 2026 **Moment Estimates and Discrepancy for Sums of Square Roots Modulo One**, FOUVRY–73 Summer School (short talk), Bernoulli Center, EPFL, Lausanne, Switzerland.

5 Aug 2026 **Moment Estimates and Discrepancy for Sums of Square Roots Modulo One**, UNSW Number Theory Seminar, UNSW Sydney.

10 Dec 2025 **Least zero of pairs of additive cubic equations**, AustMS Annual Meeting, La Trobe University, Melbourne, Australia.

9 Sep 2025 **Diophantine equations and cubic forms**, Prime Pals / Number Theory Group Seminar (internal), UNSW Canberra.

Research Experience

Research Assistant, Prof. Igor E. Shparlinski, School of Mathematics and Statistics, UNSW Sydney, 7 April 2026–present.

Conducting research in analytic number theory, including character sums, large-sieve inequalities, Kloosterman and Salié sums, and arithmetic distribution problems.

Research Assistant, Prof. Hongze Li, Shanghai Jiao Tong University, 2021–present.

Conducting research on exponential sums, Diophantine equations, cubic forms, and the Hardy–Littlewood circle method, including proof development and manuscript preparation.

Teaching

Term 2, 2026 **MATH1231 Mathematics 1B**, Course Marker, UNSW Sydney.

Term 1, 2026 **MATH2018 Engineering Mathematics 2D**, Course Marker, UNSW Sydney.

Term 1, 2026 **MATH1141 Higher Mathematics 1A**, Course Marker, UNSW Sydney.

2025–2026 **MATH8000 Student Support**, Examination Invigilation and Marking, UNSW Sydney. Term 3, 2025; Terms 1–2, 2026.

2022–2024 **Elementary Number Theory (Summer Term)**, Teaching Assistant, Shanghai Jiao Tong University.

2021–2023 **Mathematical Analysis**, Teaching Assistant, Shanghai Jiao Tong University, Fall terms.